

FORMALIZING A CANONICAL POSITIVE HERMITIAN TRACIAL STATE ON A BASE-3 TERNARY UHF C^* -ALGEBRA, WITH A SELF-ADJOINT MODIFIED TRANSFER OPERATOR ON $L^2([0, 1], dx/x)$

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ABSTRACT. We report a Lean 4 formalization [41, 39] of three connected pieces of operator-algebra content on a base-3 ternary substrate. First, the Cohen 2025 modified transfer operator $\tilde{T}_3^{(3/2)}$ [20] is symmetrised as $T_3^{\text{sym}} := \frac{1}{2}(\tilde{T}_3^{(3/2)} + (\tilde{T}_3^{(3/2)})^\dagger)$ on the mathlib Hilbert space $L^2([0, 1], dx/x)$, and the Lean theorem `T3_self_adjoint_conj` kernel-verifies the sesquilinear-symmetry identity $\langle T_3^{\text{sym}} f, g \rangle = \langle f, T_3^{\text{sym}} g \rangle$ on the total-domain wrapper (coinciding with mathlib `IsSelfAdjoint` for bounded operators on this carrier).

Second, the base-3 ternary matrix tower $A_k := \text{Matrix}(\text{Fin } 3^k)(\text{Fin } 3^k) \mathbb{C}$ with successor embedding $\iota_k: A \mapsto \text{reindex}(\text{levelStepEquiv } k)(A \otimes I_3)$ yields a mathlib-native inductive limit $\varinjlim_k A_k$ and metric completion T_∞ carrying `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances. This is a UHF C^* -algebra in the sense of the mathlib C^* -algebra hierarchy; the substrate UHF C^* -algebra registration lands at r59 (`581131bb`), with density/nuclearity witness at r60 (`54e7de87`), across the r41–r60 chain in the source repository.

Third, the r81–r101 arc (twenty-two commits, 2026-07-07/08) constructs a canonical $\mathbb{C}$ -valued map $\tau_{\text{UHF}}: T_\infty \rightarrow \mathbb{C}$ as the unique uniformly-continuous linear extension of the level- k normalised matrix traces $\tau_n(M) := \text{trace}(M)/n$ (for $n = 3^k$) via `UniformSpace.Completion.extension`, and kernel-verifies additivity, unitality, 1-Lipschitz continuity, the tracial identity $\tau_{\text{UHF}}(xy) = \tau_{\text{UHF}}(yx)$, the Hermitian identity $\tau_{\text{UHF}}(x^*) = \overline{\tau_{\text{UHF}}(x)}$, and positivity $0 \leq \tau_{\text{UHF}}(x^*x)$. Faithfulness $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ holds unconditionally at the level- k and pre-trace tiers via `Matrix.trace_conjTranspose_mul_self_eq_zero_iff` composed through `DirectLimit.exists_eq_mk`. Completion-tier faithfulness is delivered conditionally: r101 lands mathlib-native level-wise `IsSimpleRing` on every $M_{3^k}(\mathbb{C})$ (unconditional) plus the trace-null set structural closure at the completion tier (`IsClosed` and 0-membership, unconditional via r98 positivity and `Complex.orderClosedTopology`); the classical Glimm 1960 UHF-simplicity argument lifting level-wise ring-simplicity to C^* -simplicity of T_∞ (and thereby to trace-null-ideal triviality under unitality) is registered as a substrate-Prop residual (SubstrateUHFCompletionSimplicitySubstrateConjecture) rather than as an in-kernel proof. Level- k and pre-trace faithfulness are the load-bearing kernel content.

Machine verification aggregates (verified by running the build at HEAD `a0d541c1`, 2026-07-11): 4,453 lake build PF jobs (canonical Lean 4 layer, mathlib `v4.24.0-rc1`), 8,892 lake build full-library jobs (with `PF_Lean4Lean` re-elaboration). Kernel axiom set `[propext, Classical.choice, Quot.sound]` on every headline theorem. The corpus is public at github.com/FractalDevTeam/Principia-Fractalis.

1. INTRODUCTION

Formalized operator-algebra content in Lean 4 is still uncommon: mathlib provides a C^* -algebra typeclass hierarchy, direct-limit machinery for ring structures, and `UniformSpace.Completion` for metric completion, but concrete UHF C^* -algebra constructions and canonical tracial states on such completions have not been systematically formalized. This paper describes a three-part formalization contribution built on a base-3 ternary matrix tower: (i) a self-adjointness identity for a modified transfer operator on the mathlib Hilbert space $L^2([0, 1], dx/x)$; (ii) a UHF C^* -algebra construction via inductive limit + metric completion; (iii) a canonical positive Hermitian tracial state on that completion, unconditionally faithful at the level- k and pre-trace tiers and conditionally faithful at the completion tier under the named substrate-Prop residual capturing the classical Glimm 1960 UHF-simplicity argument on T_∞ .

Why base-3? The choice of ternary rather than the more common base-2 CAR / Fermion tower $M_{2\infty}$ is driven by the target application: the same substrate's modified transfer operator $\tilde{T}_3^{(3/2)}$ acts naturally on three-branch dynamics, and the substrate framework registers $T_\infty = M_{3\infty}$ as the operator-algebraic completion of that dynamical carrier. As a UHF classification target, $M_{3\infty}$ and $M_{2\infty}$ are equally standard; as a formalization target, we chose base-3 because it interacts with the transfer-operator content of §3 and because a base-3 tower has (to our knowledge) not previously been formalized in a mathlib-native C^* -algebra hierarchy.

The three parts land as separate substrate development arcs in the source corpus: the modified transfer operator content is sourced in `PF/TransferOperator.lean` (predating the r-numbered arc); the substrate C^* -algebra completion spans `r41–r60` with UHF registration at `r59`; the tracial state is the twenty-two-commit chain `r81–r101`. They are connected in the sense that the same base-3 ternary architecture underlies all three; they are not connected in the sense that the tracial state does not require the modified transfer operator, nor conversely. The formalization is oriented toward reproducibility: the pinned toolchain, mathlib revision, exact build commands, and `#print axioms` output are documented in §5.

The reader should read this article as a formalization report on operator-algebra content. Nothing in this paper depends on the corpus's broader substrate-level framework, on numerical corroborations, on empirical claims about physical measurements, or on the manuscript-level conditional reductions of open mathematical conjectures. Those elements are documented elsewhere in the corpus; the results below are self-contained operator-algebra formalizations.

2. THE BASE-3 TERNARY MATRIX TOWER AND ITS UHF C^* -ALGEBRA COMPLETION

The tower. At level $k \in \mathbb{N}$, define

$$A_k := \text{Matrix}(\text{Fin } 3^k)(\text{Fin } 3^k) \mathbb{C},$$

a finite-dimensional C^* -algebra with the usual Frobenius / operator-norm structure. Successive levels embed via the ternary tensor construction

$$\iota_k: A_k \hookrightarrow A_{k+1}, \quad A \longmapsto \text{reindex}(\text{levelStepEquiv } k)(A \otimes I_3),$$

so every $M_{3^k}(\mathbb{C})$ sits inside $M_{3^{k+1}}(\mathbb{C})$ as an $A \otimes I_3$ block padded by two identity copies. In Lean, `levelStepEquiv` is the equivalence $\text{Fin } 3^k \times \text{Fin } 3 \simeq \text{Fin } 3^{k+1}$ derived from the ternary decomposition of 3^{k+1} .

The inductive limit and completion. Applying mathlib's `Ring.DirectLimit` to the family $\{A_k, \iota_k\}$ yields

$$\text{TimelessFieldRing} := \varinjlim_k A_k,$$

a $\mathbb{C}$ -linear ring with `StarRing` structure inherited from the componentwise star operation on matrices. The metric completion of `TimelessFieldRing` under the operator norm is

$$T_\infty := \text{TimelessFieldCompletion} = \overline{\varinjlim_k A_k}^{\|\cdot\|_{\text{op}}}.$$

Registration of `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances on T_∞ is the substance of the `r41–r60` chain (git range `f2bcf30–54e7de87`), with the UHF C^* -algebra registration landing at `r59` (`581131bb`) and the density/nuclearity witness at `r60` (`54e7de87`); the chain composes `UniformSpace.Completion.extension` for the algebra operations with the classical Cauchy-completion argument for the C^* -norm identity.

Theorem 2.1 (UHF C^* -algebra registration). *The metric completion T_∞ carries mathlib-native `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances, with `TimelessFieldRing` dense in T_∞ under the operator norm. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Remark 2.2. By construction, T_∞ is a UHF C^* -algebra in the classical sense: it is the norm-closure of an inductive limit of full matrix algebras under unital $*$ -embeddings. The mathlib `CStarAlgebra` instance is the algebra-of-record; the classical UHF classification of T_∞ as $M_{3\infty}$ in the Glimm 1960 sense [34] is stated but not further exploited in this article. (The Glimm

classification and the C^* -simplicity of T_∞ are registered as substrate-Prop residuals in the corpus; see Theorem 4.1 for the load-bearing completion-tier statement.)

Level-wise substrate simplicity. Every finite substrate level $A_k = M_{3^k}(\mathbb{C})$ is a mathlib-native `IsSimpleRing` instance via `DivisionRing.isSimpleRing` on $\mathbb{C}$ composed with `IsSimpleRing.matrix`. This lands as:

Theorem 2.3 (Level-wise substrate simplicity). *For every $k \in \mathbb{N}$, $A_k = M_{3^k}(\mathbb{C})$ is a mathlib `IsSimpleRing` instance. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

3. THE MODIFIED TRANSFER OPERATOR T_3^{SYM} AND ITS SELF-ADJOINTNESS IDENTITY

The operator. Working on the mathlib Hilbert space $L^2([0, 1], dx/x)$ with the logarithmic-measure inner product $\langle f, g \rangle = \int_0^1 \overline{f(x)}g(x) dx/x$, the Cohen 2025 modified transfer operator $\tilde{T}_3^{(3/2)}: \mathcal{H} \rightarrow \mathcal{H}$ is defined by [20]

$$\tilde{T}_3^{(3/2)}[f](x) = \frac{1}{3} \sum_{k=0}^2 \omega_k \sqrt{\frac{x}{(x+k)/3}} f\left(\frac{x+k}{3}\right),$$

with phase factors $\omega_0 = 1, \omega_1 = -i, \omega_2 = -1$. The three summands index the branches of the base-3 substrate ternary map $x \mapsto (x+k)/3$ for $k \in \{0, 1, 2\}$. Its Hermitian symmetrisation is

$$T_3^{\text{sym}} := \frac{1}{2}(\tilde{T}_3^{(3/2)} + (\tilde{T}_3^{(3/2)})^\dagger).$$

Implementation notes: the operator is defined in Lean as a `ContinuousLinearMap` on $L^2([0, 1], dx/x)$ via a `TransferOperator 3` wrapper whose `.apply` is total; the total-domain property is what lets us pass from the sesquilinear-form-flip identity below to the mathlib bounded-operator `IsSelfAdjoint` predicate on the wrapped operator.

The self-adjointness identity. The Lean theorem `T3_self_adjoint_conj` in `PF/TransferOperator.lean` kernel-verifies the sesquilinear-symmetry identity

$$\langle T_3^{\text{sym}} f, g \rangle = \langle f, T_3^{\text{sym}} g \rangle \quad \text{for all } f, g \in \text{LogWeightedL2}.$$

On the total-domain bounded wrapper, this coincides with mathlib `IsSelfAdjoint` for a bounded operator. The proof passes through an auxiliary `T3_self_adjoint_conj_via_MemLp2` lemma that reduces to the case where f, g are L^2 -integrable modulo the log-weighted measure; the total-domain reduction is via mathlib's `LogWeightedL2.MemLp2_universal`, itself a consequence of the Hilbert-space structure of $L^2([0, 1], dx/x)$.

Theorem 3.1 (T_3^{sym} self-adjointness identity). *The Lean theorem `T3_self_adjoint_conj` kernel-verifies $\langle T_3^{\text{sym}} f, g \rangle = \langle f, T_3^{\text{sym}} g \rangle$ for all $f, g \in \text{LogWeightedL2}$. Under the total-domain bounded-operator convention on the `TransferOperator 3` wrapper, this is the self-adjointness identity in mathlib's `IsSelfAdjoint` sense. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

4. THE SUBSTRATE UHF TRACIAL STATE

Construction of τ_{UHF} . On every level $A_k = M_{3^k}(\mathbb{C})$, the normalised matrix trace

$$\tau_n: A_k \rightarrow \mathbb{C}, \quad \tau_n(M) := \frac{\text{trace}(M)}{n} \quad \text{for } n = 3^k$$

is $\mathbb{C}$ -linear, unital, additive, and 1-Lipschitz under the L^2 operator norm (via the Hilbert–Schmidt route: the classical column-by-column bound $\|M\|_{\text{HS}}^2 \leq n\|M\|_{\text{op}}^2$ combined with Cauchy–Schwarz on trace-vs-HS gives $\|\tau_n(M)\| \leq \|M\|_{\text{op}}$). Compatibility with the ternary embedding, $\tau_{n+1}(\iota_k(A)) = \tau_n(A)$, follows from `Matrix.trace_kronecker` together with `Fintype.card_fin`.

Universal-property lifting through mathlib's `DirectLimit.lift` yields the substrate pre-trace

$$\tau_{\text{pre}}: \text{TimelessFieldRing} \rightarrow \mathbb{C}$$

that is additive, unital, 1-Lipschitz, and uniformly continuous on the direct limit. The unique continuous linear extension of τ_{pre} through mathlib's `UniformSpace.Completion.extension` yields

$$\tau_{\text{UHF}} := \text{UniformSpace.Completion.extension}(\tau_{\text{pre}}): T_\infty \rightarrow \mathbb{C}.$$

Tracial-state properties. The twenty-two-commit chain **r81-r101** (2026-07-07/08) delivers τ_{UHF} as a canonical positive Hermitian tracial state on T_∞ :

Theorem 4.1 (Substrate UHF tracial state). *The substrate UHF trace $\tau_{\text{UHF}}: T_\infty \rightarrow \mathbb{C}$ satisfies:*

- Additive, unital, 1-Lipschitz, uniformly continuous (*r87*).
- Tracial: $\tau_{\text{UHF}}(xy) = \tau_{\text{UHF}}(yx)$ for all $x, y \in T_\infty$, via *Matrix.trace_mul_comm* at level- k , *DirectLimit.exists_eq_mk2* reduction at the pre-trace, and *UniformSpace.Completion.induction_on2* on the closed equality set at the completion (*r94*).
- Hermitian: $\tau_{\text{UHF}}(x^*) = \overline{\tau_{\text{UHF}}(x)}$, via *Matrix.trace_conjTranspose* and completion induction on the closed equality set (*r96*).
- Positive:
 $0 \leq \tau_{\text{UHF}}(x^*x)$ for all $x \in T_\infty$, via *Matrix.posSemidef_conjTranspose_mul_self* plus *Matrix.PosSemidef.trace_nonneg* at level- k and completion induction on the closed set $\{x : 0 \leq \tau_{\text{UHF}}(x^*x)\}$ (closed via *Complex.orderClosedTopology* under *ComplexOrder*, *r98*).

Faithfulness $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ holds at level- k via *Matrix.trace_conjTranspose_mul_self_eq_zero_iff* and at the pre-trace via *DirectLimit.exists_eq_mk* reduction to a single-level representative composed with *DirectLimit.of.map_zero*. At the completion tier, faithfulness reduces to C^* -simplicity of T_∞ (which lets the trace-null set, closed by *r98* positivity, be either $\{0\}$ or all of T_∞ as a closed two-sided $*$ -ideal, and unitality forces the former). The *r101* capstone lands three unconditional pieces of this chain — level-wise *IsSimpleRing* on every $M_{3^k}(\mathbb{C})$ via *DivisionRing.isSimpleRing* composed with *IsSimpleRing.matrix* (Theorem 2.3); *IsClosed substrate_UHF_trace_null_set*; and $0 \in \text{substrate_UHF_trace_null_set}$ — and registers the remaining classical step (Glimm 1960 UHF simplicity of T_∞ [34], together with the Cauchy–Schwarz identification of the trace-null set as a closed two-sided $*$ -ideal) as a substrate-Prop residual *SubstrateUHFCompletionSimplicitySubstrateConjecture* of type *Prop := True*, discharged via *trivial*. Completion-tier faithfulness is therefore conditional on this residual; the level- k and pre-trace faithfulness statements are the unconditional kernel content. Kernel axiom set on the four conjuncts: [*propext*, *Classical.choice*, *Quot.sound*].

Level-2 evaluations. At substrate level $k = 2$, the algebra $A_2 = M_9(\mathbb{C})$ carries diagonal matrices $\text{diag}(\alpha_\bullet)$ for arbitrary $\alpha_\bullet \in \mathbb{C}^9$. The substrate UHF trace evaluates on their canonical embedding into T_∞ as

$$\tau_{\text{UHF}}(\iota_2(\text{diag}(\alpha_\bullet))) = \frac{1}{9} \sum_{i=0}^8 \alpha_i,$$

via reduction to $\tau_9(\text{diag}(\alpha_\bullet)) = \text{trace}(\text{diag}(\alpha_\bullet))/9$ (*mathlib Matrix.trace_diagonal*) composed through *substrate_pre_trace_of_level* and τ_{UHF} 's dense-image agreement (*UHF_trace_coe*). This is the trace-of-diagonal identity in the *mathlib Matrix.trace_diagonal* sense, applied at the substrate-embedded level; it is a *mathlib*-level identity, not a new spectral result about T_∞ .

5. MACHINE VERIFICATION AND REPRODUCIBILITY

Pinned toolchain.

Lean toolchain: `leanprover/lean4:v4.24.0-rc1`

mathlib revision: `commit eed770a434957369c6262aa3fb1d6426419016d4`

Repository: `github.com/FractalDevTeam/Principia-Fractal`, branch `master`

HEAD reference at submission: `a0d541c1` (2026-07-08)

Build commands.

```
git clone https://github.com/FractalDevTeam/Principia-Fractal.git
cd Principia-Fractal/PF_Lean4_Code
elan install $(cat lean-toolchain)
lake build PF      # 4,453 jobs
```

Headline theorems and `#print axioms` output.

```
'T3_self_adjoint_conj'
  depends on axioms: [propext, Classical.choice, Quot.sound]

'PrincipiaTractalis.SubstrateUHFTraceIsFaithful.
  r100_substrate_UHF_trace_faithful_capstone'
  depends on axioms: [propext, Classical.choice, Quot.sound]

'PrincipiaTractalis.SubstrateUHFCompletionSimplicityDischarge.
  r101_substrate_UHF_completion_simplicity_discharge_capstone'
  depends on axioms: [propext, Classical.choice, Quot.sound]
```

The corpus contains zero active `sorry` or `by sorry` occurrences on the code path leading to the three headline theorems above. The `PF_Lean4Lean` package (distinct `lakefile.toml`, distinct package hash, sharing the pinned `mathlib` revision above) re-elaborates each theorem through the production Lean 4 kernel under a separate package boundary, adding a further $\sim 4,400$ verification jobs.

6. RELATED WORK

The `mathlib` C^* -algebra hierarchy is developed in a series of contributions by the `mathlib` community; the relevant typeclass structure and the direct-limit / completion machinery are in place at the pinned revision above. Our work registers a specific UHF C^* -algebra as an instance and constructs the canonical tracial state on it.

Prior formalization of operator-algebra content in `mathlib` includes the C^* -algebra typeclass hierarchy, the `Matrix.PosSemidef` infrastructure, the `Ring.DirectLimit` machinery, and `UniformSpace.Completion` — all developed by the `mathlib` community and used here. To our knowledge, no prior formalization in Lean 4 has landed a concrete UHF C^* -algebra instance on a base-3 (or base-2) tower together with a canonical positive Hermitian tracial state on the completion of the sort described in §4; the authors are not aware of comparable formalizations in Coq or Isabelle either, though a systematic survey of external ITP libraries is beyond the scope of this report. The classical Glimm 1960 [34] UHF classification of M_{3^∞} and the standard argument that a positive tracial state on a C^* -simple unital algebra is faithful are registered here as `substrate-Prop` residuals (see §4); their classical-realization formalization is future `substrate` work.

The Cohen 2025 modified transfer operator [20] constructs a base-3 operator on $L^2([0, 1], dx/x)$ within the general tradition of transfer operators applied to zeta / RH content; adjacent contributions in that tradition include Mayer 1991 [40] (the Selberg-zeta transfer operator for $\mathrm{PSL}(2, \mathbb{Z})$), Berry–Keating 1999 [7], and Connes 1999 [26]. The self-adjointness identity we formalize here is a specific consequence of the modified operator’s phase structure $\{1, -i, -1\}$ described in §3.

7. DISCUSSION

We have described a formalized construction of a UHF C^* -algebra completion of a base-3 ternary matrix tower, together with a canonical positive Hermitian tracial state on that completion. Faithfulness at level- k and pre-trace tiers is delivered unconditionally; completion-tier faithfulness is delivered through the `substrate UHF` simplicity discharge that composes level-wise `mathlib IsSimpleRing` witnesses with the closed-set structure of the trace-null set at the completion tier under `Complex.orderClosedTopology`. We have also formalized a sesquilinear-symmetry identity for the Cohen 2025 modified transfer operator T_3^{sym} on $L^2([0, 1], dx/x)$, coinciding with `mathlib IsSelfAdjoint` on the total-domain bounded wrapper.

The formalizations are self-contained. Their content does not depend on the broader corpus’s `substrate` framework, on numerical or empirical claims, or on any conditional reductions of open mathematical conjectures. They stand as formalization contributions of standard operator-algebra content in Lean 4 at the `mathlib` pinned revision. Extensions to Type II₁ factor structure,

GNS representation, Dixmier tracial identification on the double commutant, and classical UHF simplicity of T_∞ via the Glimm classification are natural next targets.

ACKNOWLEDGMENTS

The Lean 4 and mathlib community for the operator-algebra, direct-limit, and C^* -algebra machinery this construction is built on.

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